

Context-Aware, Inspectable Travel Optimization

Updated literature synthesis, research gap, and first-paper scope

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Meeting with Prof. Choi

Core question: how can we connect **traveler intent**, **formal feasibility**, **context-aware repair**, and **user correction**?

Start with one realistic request

Example request

“I want a relaxed seven-day California trip. Yosemite is mandatory. I prefer nature, but I do not want strenuous hiking or more than three hours of driving per day.”

LLM-only risk

- ▶ Fluent but infeasible
- ▶ Wrong opening hours or travel order
- ▶ Hidden interpretation errors

Solver-only risk

- ▶ Requires manual formalization
- ▶ User intent becomes opaque weights
- ▶ Feasible does not mean desirable

Opaque hybrid risk

- ▶ Correct JSON, wrong meaning
- ▶ No way to inspect assumptions
- ▶ No faithful why-not explanation

The research problem is not merely generating a route; it is preserving a traceable chain from language and context to a feasible, useful, and correctable itinerary.

A travel plan can fail at six different levels



valid JSON $\nrightarrow$ correct interpretation $\nrightarrow$ feasible route $\nrightarrow$ good itinerary.

Automatic checks are strongest for
Schema, entity IDs, units, time windows,
route continuity, budget, and solver status.

Human confirmation remains
important for

Must-go meaning, hard versus soft
preferences, vague language, consequential
repairs, and unstated needs.

Logic-LM; TTG; TravelPlanner; TravelEval; TripScore

Literature map: organize by the failure being solved

Planning challenge	What prior work contributes	What still remains
Attractive POI selection	Personalized recommendation, user utility, context-aware scores	Attractive places can still be infeasible or poorly sequenced
Feasible scheduling	OP / TTDP, exact optimization, branch-and-check	Solver conflicts are usually hidden from travelers
Natural-language input	Logic-LM, TRIP-PAL, TTG, LLMAP	Executable structure may still misrepresent intent
Complete-plan evaluation	TravelEval and TripScore	Feasibility alone does not establish practical quality
Disruption response	TripTide and adaptive planning	Revisions may unnecessarily destroy original intent
User understanding	Mixed initiative, XAI, visual analytics, CityHood	Explanations are often post-hoc, ungrounded, or not tied to correction

Combined gap: prior work usually optimizes, interprets, evaluates, repairs, or explains one part of the pipeline; less work connects all parts through auditable artifacts and user correction.

TTDP foundation: recommendation is not the same as planning

Recommendation question

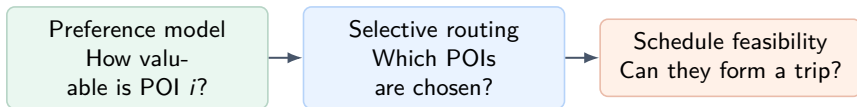
Which POIs would this traveler value?

- ▶ Interest and category fit
- ▶ Popularity and social signals
- ▶ Weather- or context-adjusted suitability

Planning question

Which valued POIs can be visited, and in what order?

- ▶ Travel and visit time
- ▶ Opening windows and daily limits
- ▶ Budget, hotels, bases, mandatory visits



Basic POI selection, sequencing, time windows, and personalized reward are established foundations; they should not be claimed as the primary novelty.

Vansteenwegen et al. 2011; Ruiz-Meza and Montoya-Torres 2022; Halder et al. 2024

Rich constraints: four branch-and-check / solve variants

Method	Question asked	What the algorithm learns
B-B&C	Can all selected POIs be scheduled?	Yes/no feasibility; simple cut excludes the full incompatible set
B-B&S	What is the best feasible subset?	Salvages a good itinerary from an infeasible candidate bundle
E-B&C	Can we reject conflicts earlier and more strongly?	Smaller conflict sets, incompatibility cuts, duration bounds, partial-node pruning
E-B&S	What is the best subset with enhanced search?	Strong incumbent plus stronger conflict information, at higher subproblem cost

Example solver conflict

Yosemite Valley + Mariposa Grove + Fresno evening activity exceed the available day by 96 minutes.

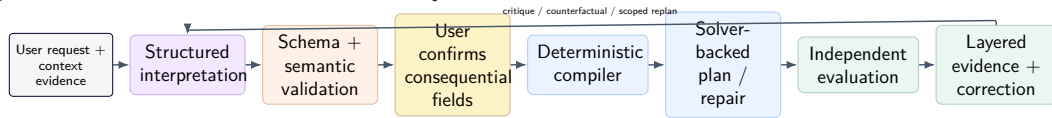
Possible user-visible repairs

Extend the day, move Fresno, remove one stop, change the base, or relax a soft preference.

The same conflict evidence that improves solver search can support a faithful **why-skipped** or **what-would-change** explanation.

Vu et al. 2022, Branch-and-Check Approaches for TTDP with Rich Constraints

Logic-LM: formulate, solve, interpret, refine



Machine-owned responsibilities

Grounding, deterministic compilation, modeled feasibility, optimization, independent checking, and artifact lineage.

Human-owned responsibilities

Confirm ambiguous meaning, correct preferences, approve consequential scope changes, and choose among eligible alternatives.

This is a **formulate–validate–solve–evaluate–explain–correct** loop, not an autonomous LLM route agent.

Strong idea

The solver is both a reasoning engine and a debugger for malformed formalizations.

Critical limitation

A representation may execute successfully while encoding the wrong meaning.

Solver feedback can repair low-consequence syntax and grounding errors; semantic or outcome-sensitive changes still require confirmation.

TTG and LLMAP: language should guide the solver, not replace it

TTG: natural language to MILP

- ▶ Fine-tuned translator produces structured JSON
- ▶ Constrained decoding guarantees valid structure
- ▶ MILP produces a feasible / optimal solution for the encoded request
- ▶ Translation errors can still degrade the final plan

Ju et al. 2024, To the Globe

LLMAP: LLM-as-parser

- ▶ Parses POI types, dependencies, deadlines, and priorities
- ▶ Graph search handles POI selection and routing
- ▶ Supports conversational correction
- ▶ LLM preference weights are broad and step-like, not finely calibrated

Yuan et al. 2025, LLMAP

Avoid false precision

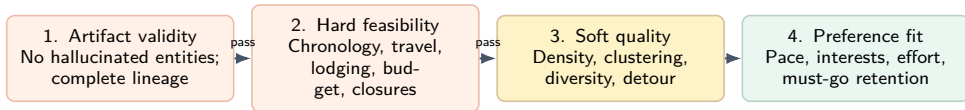
`nature_weight = 0.783`

Prefer interpretable levels

Nature importance: **high**
Driving sensitivity: **medium**

The compiler—not the LLM—should map confirmed semantic levels into versioned objective ranges and constraints.

TravelEval and TripScore: feasible is not the same as good



TravelEval contribution

Complete-plan evaluation across accuracy, compliance, temporality, spatiality, economy, and utility.

TripScore contribution

Hard gates plus soft and preference metrics, combined into an expert-calibrated reward.

An infeasible itinerary should never receive a high score merely because it contains attractive landmarks. Report component metrics even when a unified reward is used.

Chen et al. 2026, TravelEval; Qu et al. 2025, TripScore

TripTide: adaptation should preserve intent, not restart blindly

Disruption scope

- ▶ Stop-level
- ▶ Day-level
- ▶ Plan-level

Traveler tolerance

- ▶ Flexible traveler
- ▶ Change-averse traveler
- ▶ Explicit repair scope

Adaptation outcomes

- ▶ Responsiveness
- ▶ Intent preservation
- ▶ Semantic / spatial / sequence change

Optimization view

$$\max \quad U(\text{repair}) - \lambda_C \text{ context burden} - \lambda_T \text{ travel burden} - \lambda_\Delta \Delta(\text{original plan})$$

Weather is a useful first disruption case, but the core research mechanism is **minimal-change, evidence-grounded repair**.

Explainability: explain the input, process, and output

Input explanation

- ▶ What does the system think I prefer?
- ▶ Which fields are hard or soft?
- ▶ Which interpretations are uncertain?

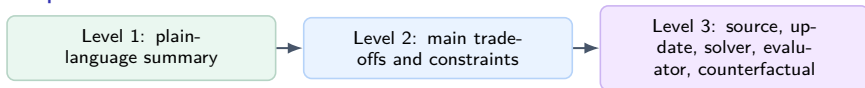
Process explanation

- ▶ Which evidence changed the model?
- ▶ Which constraints were binding?
- ▶ Solver, fallback, and evaluator status?

Output explanation

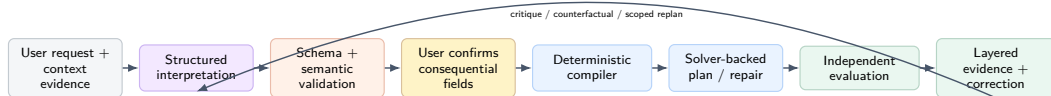
- ▶ Why selected or skipped?
- ▶ What changed after disruption?
- ▶ What would need to change?

Layered explanation



Choose visual form from the user task: map for spatial change, timeline for feasibility, bars for tradeoffs, and side-by-side diff for repair.

Combined architecture emerging from the literature



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Confirm ambiguous meaning, correct preferences, approve consequential scope changes, and choose among eligible alternatives.

This is a **formulate–validate–solve–evaluate–explain–correct** loop, not an autonomous LLM route agent.

Guarantee boundary: what “solver-backed” really means

The solver can establish

- ▶ Feasibility under encoded constraints
- ▶ Optimality or a reported optimality gap
- ▶ Infeasibility of the encoded model
- ▶ Effects of declared objective parameters

The solver cannot establish

- ▶ That language was interpreted correctly
- ▶ That provider data are complete or current
- ▶ That all human needs are modeled
- ▶ That this is the globally best real-world trip

Permitted claim

Given the **user-confirmed structured interpretation**, **frozen candidate and context data**, and **declared mathematical model**, the solver can certify modeled feasibility or optimality for that model when the backend returns the corresponding certificate.

Solver-certified $\neq$ automatically real-world valid.

Current project maturity: implemented, partial, and proposed

Implemented but not fully validated	Partially implemented	Proposed
Multi-day California route artifacts	Interest profiles and structured overrides	Natural-language parser and semantic confirmation
Hotels, base cities, nature regions	Weather-aware scoring and constraints	Deterministic parser-to-planner compiler
Gurobi and heuristic/fallback routes	Provider metadata and source confidence	Immutable PlannerRun / Plan / Evaluation lineage
Route alternatives and dashboard views	Basic why-selected and some why-not codes	Independent complete-plan evaluator
Existing weather, NPS, OSM, Wikidata, OSRM paths	Solver/fallback fields in scattered outputs	Minimal-change repair and branch/check conflicts
	Browser-side preference preview	Controlled inspectability / correction study

Important blocker: a route changed after solving cannot retain the original solver certification; it needs new run lineage and complete evaluation.

Novelty matrix: what exists, and what we can contribute

Capability	Prior work already establishes	Project opportunity
Constrained itinerary planning	OP / TTDP, rich constraints, exact and heuristic methods	Weather- and nature-aware multi-day repair with truthful solver/fallback lineage
Language-to-symbolic input	Logic-LM, TRIP-PAL, TTG, LLMAP	User-visible interpretation, confirmation, and deterministic compilation
Complete-plan evaluation	TravelEval, TripScore	Independent evaluator including context, repair scope, provenance, and explanation grounding
Disruption response	TripTide and adaptive benchmarks	Formal minimal-change repair with conflict-based alternatives
Visual explanation	Recommender XAI, visual analytics, CityHood	Evidence chain from request and context to update, solver result, and plan difference

Defensible combined contribution

An inspectable solver-backed workflow that converts confirmed travel intent and context into minimal-change repairs, independently evaluates the plans, and exposes grounded evidence for correction.

“LLM + solver” by itself is not the novelty.

First publishable slice versus long-term roadmap

First research slice

- ▶ Truthful immutable run and plan artifacts
- ▶ Structured or gold user-confirmed requests
- ▶ Weather, closures, travel-time disruption
- ▶ Traveler-approved pace / accessibility
- ▶ Context-blind, heuristic, and solver repair baselines
- ▶ Original-versus-repaired metrics
- ▶ Independent gated evaluation
- ▶ California; second corridor only if reproducible

Longer-term system

- ▶ Validated LLM request parser
- ▶ Conversational correction and deferral
- ▶ Enhanced branch-and-check / solve
- ▶ Wider context-family provider registry
- ▶ Nationwide coverage audit
- ▶ Proactive monitoring and notification
- ▶ User-authorized adoption or booking safeguards

Narrowing the first paper is not abandoning the larger vision; it creates a coherent scientific story that can be validated before autonomy is added.

Research questions and evidence map

Research question	Core comparison	Main evidence
RQ1: Interpretation	Direct LLM JSON vs. validated / confirmed interpretation	Schema validity, field accuracy, hard-soft classification, dependency direction, confirmation / deferral
RQ2: Planning and repair	Context-blind vs. heuristic vs. solver-backed repair vs. full replanning	Feasibility, utility, context burden, travel, cost, intent retention, edit distance, runtime, gap
RQ3: Evaluation	Planner totals vs. independent gated evaluator	Violation detection, pairwise expert agreement, rank stability, component uncertainty
RQ4: Inspectability	Basic interface vs. layered evidence and counterfactuals	Detection, correction accuracy, revision quality, time, workload, perceived control, calibrated reliance
RQ5: Action policy	Always act vs. always confirm vs. consequence-aware policy	Unsafe auto-repair, unnecessary confirmation, escalation, and deferral accuracy

The first paper should answer a coherent subset of these RQs; it should not claim the full pipeline without corresponding implementation and evidence.

Venue paths require different evidence

	TRB path	IUI or CHI path
Central claim	Context-aware transportation planning and minimal-change disruption repair	Mixed-initiative interpretation, correction, explanation, and calibrated reliance
Primary evidence	Frozen scenarios, route baselines, robustness, ablations, operational error analysis	Parser / confirmation evaluation, controlled interaction study, behavioral outcomes
Human evidence	Expert review supports operational validity	Ethics-approved participants required for user-benefit claims
Parser requirement	Optional for first technical paper	Required for interpretation claims
Main risk	Interface-heavy story may appear insufficiently transportation-centered	Dashboard demo without controlled evidence is insufficient

Select the venue after checking which mechanism is implemented and which evidence is complete; do not relabel one unfinished manuscript.

Decisions needed from this meeting

1. Should the first paper center on **solver-backed context repair** or **validated interpretation and correction**?
2. Which two or three research questions are realistic for the first study?
3. Is baseline conflict extraction needed now, or should branch-and-check remain a staged extension?
4. What human evidence is realistic: expert inspection, mixed-initiative study, or neither in the first submission?

Recommended near-term focus

Truthful artifacts + independent evaluation + weather/closure repair + original-versus-repaired explanations

Long-term agent layer: monitor disruptions, propose a scoped repair, explain the evidence, and wait for user authorization.